

# Curriculum Vitae – Danushka BOLLEGALA

## PERSONAL DETAILS

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NAME: Professor Danushka Bollegala  
POSITION: Professor in Natural Language Processing  
ADDRESS: Department of Computer Science, University of Liverpool, L69 3BX, UK.  
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## HIGHER EDUCATION

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|-------------|-----------------------------------------------------------------------------------|
| 2007 – 2009 | PhD Computer Science, The University of Tokyo, Japan.<br><i>Summa Cum Laude</i>   |
| 2005 – 2007 | M.Sc. Computer Science, The University of Tokyo, Japan.<br><i>Summa Cum Laude</i> |
| 2001– 2005  | B.Sc. Computer Science, The University of Tokyo, Japan.<br><i>Summa Cum Laude</i> |

## HONOURS AND AWARDS

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1. EPSRC Top Peer Reviewer's Award 2017.
2. Best Journal Paper of the Year 2014-2015, Japanese Society for Artificial Intelligence.
3. IEEE Young Author Award 2011.
4. Best paper award at the 2011 Genetic and Evolutionary Computation (GECCO) Conference.
5. Best poster paper at the 2010 Pacific Rim International Conference on Artificial Intelligence (PRICAI).
6. Annual Conference Award for the Best Paper at 2010 Japanese Society for Artificial Intelligence (JSAI).
7. Dean's Award for the Best Doctoral Thesis of the Year 2010, University of Tokyo.
8. Dean's Award for the Best Masters Thesis of the Year 2007, University of Tokyo.
9. Dean's Award for the Best Undergraduate Thesis of the Year 2005, University of Tokyo.

## EMPLOYMENT RECORD

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|--------------------------------|-----------------------------------------------------------------------------------------------------------|
| <i>Current</i><br>OCTOBER 2018 | Professor, UNIVERSITY OF LIVERPOOL, UK.                                                                   |
| <i>Current</i><br>JANUARY 2019 | Amazon Scholar, AMAZON, Cambridge, UK.                                                                    |
| SEPTEMBER 2013 – DECEMBER 2018 | Senior Lecturer, UNIVERSITY OF LIVERPOOL, UK.                                                             |
| APRIL 2012 – AUGUST 2013       | Senior Assistant Professor, ( <i>koshi</i> ) UNIVERSITY OF TOKYO, Japan.                                  |
| APRIL 2010 – MARCH 2012        | Assistant Professor, UNIVERSITY OF TOKYO, Japan.                                                          |
| OCTOBER 2009 – MARCH 2010      | JSPS Post-doctoral Research Fellow, UNIVERSITY OF SUSSEX, UK.                                             |
| APRIL 2007 – SEPTEMBER 2010    | Japan Society for Promotion of Science (JSPS), Doctoral Research Fellow (DC1) UNIVERSITY OF TOKYO, Japan. |
| APRIL 2005 – MARCH 2007        | Research Assistant, National Institute of Advanced Industrial Science and Technology (AIST), Japan.       |
| JUNE 2005 – MARCH 2010         | Research Consultant, FAST a Microsoft Subsidiary (former FAST Search & Transfer), Norway.                 |

## FELLOWSHIPS AND AWARDS

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|                 |                                                                                                         |
|-----------------|---------------------------------------------------------------------------------------------------------|
| 2026.04-CURRENT | Advisor Sekisui Chemicals, Japan, Vision Action Language Models for Chemical robotic manipulation.      |
| 2021.09-CURRENT | Advisor Deloitte, Japan, Medical Information Extraction Systems                                         |
| 2018.04-2019.10 | Advisor LexSnap UK, specialising in Legal Chatbot Systems                                               |
| 2017.03-2017.06 | Advisor Skwile UK, specialising in Financial Risk Prediction                                            |
| 2011.04-2011.06 | Visiting Research Fellow, Department of Computer Science, University of Cambridge, UK.                  |
| 2009.10-2010.03 | Japan Society for Promotion of Science (JSPS), Post-doctoral Research Fellow, University of Sussex, UK. |
| 2007.04-2009.09 | Japan Society for Promotion of Science (JSPS), Doctoral Research Fellow, University of Tokyo, Japan.    |
| 2000.04-2007.03 | Japan Ministry of Education Overseas Full Scholarship, University of Tokyo, Japan                       |

## LEADERSHIP, PROFESSIONAL AND COLLEGIAL EXPERIENCE

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|                 |                                                                                              |
|-----------------|----------------------------------------------------------------------------------------------|
| 2017.10-CURRENT | Head of the Machine Learning Research Group.                                                 |
| 2016.01-CURRENT | Undergraduate admissions tutor, Department of Computer Science, University of Liverpool      |
| 2013.09-2016.01 | Disability Support Officer, Department of Computer Science, University of Liverpool          |
| 2014-CURRENT    | Leader Natural Language Processing Group, University of Liverpool                            |
| 2017-2018       | Member of the Advisory Committee, AI and Future Jobs, Royal Society of Science.              |
| 2017-CURRENT    | Full member of Engineering and Physical Science Research Council (EPSRC) Peer Review College |
| 2017-CURRENT    | Assessor for the Irish Research Council                                                      |
| 2013-2015       | Associate Editor of the Transactions of the Japanese Society for Artificial Intelligence     |
| 2016-CURRENT    | Associate Editor of the Journal of Computational Social Sciences                             |
| 2014-CURRENT    | Evaluator of Research Grants for Xi'an Jiaotong-Liverpool University                         |

## ORGANISATION OF SCIENTIFIC MEETINGS

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|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026 | <b>Co-organiser</b> Third NLP for Social Good Workshop at IJCAI-EACI 2026                                                                                                                               |
| 2024 | <b>Program Chair</b> *SEM 2024 (13th Joint Conference on Lexical and Computational Semantics) at NAACL 2024 in Mexico City, Mexico                                                                      |
| 2024 | <b>Co-organiser</b> Second NLP workshop for Social Good Symposium at University of Liverpool                                                                                                            |
| 2023 | <b>Co-organiser</b> First NLP workshop for Social Good Symposium at University of Liverpool                                                                                                             |
| 2018 | <b>Co-organiser</b> of the Knowledge Representation and Reasoning in Natural Languages (KRNL) Workshop at the 16th International Conference on the Principles of Knowledge Representation and Reasoning |
| 2017 | <b>Local organiser</b> for the 17th Annual Meeting of the International Society of Pharmacovigilance (ISoP)                                                                                             |
| 2017 | <b>Program chair</b> of the Pharmacovigilance and Social Media Workshop at ISoP                                                                                                                         |
| 2012 | <b>Program chair</b> of the International Organised Sessions (IOS) at the 26th Annual Conference of Japanese Society for Artificial Intelligence                                                        |

## PROGRAM COMMITTEES

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|--------|-------------------------------------------------------------------------------------------------------------------------------|
| 2026   | Senior Area Chair, EMNLP-2026                                                                                                 |
| 2026   | Senior Area Chair, ACL-2026                                                                                                   |
| 2025   | Senior Area Chair, EMNLP-2024                                                                                                 |
| 2024   | Senior Area Chair on Ethics and Social Biases at the 31st International Conference on Computational Linguistics (COLING-2025) |
| 2024   | Program Chair of the 13th Joint Conference on Lexical and Computational Semantics (*SEM 2024)                                 |
| 2023   | Program Chair of the ACL 2023 System Demonstrations                                                                           |
| 2021   | Action Editor of the ACL Rolling Review System                                                                                |
| 2021   | Area Chair of the Interpretability Track at ACL-2021                                                                          |
| 2019   | Area Chair of the Machine Learning Track at EMNLP-2019                                                                        |
| 2014 – | Senior Programme Committee member for the International Joint Conference in Artificial Intelligence                           |
| 2014–  | Senior Programme Committee member for the AAAI Conference on Artificial Intelligence                                          |
| 2014   | Information Extraction Area Chair for the International Conference on Computational Linguistics (COLING)                      |
| 2010–  | PC member of ACL, EMNLP, NIPS, WWW, COLING, LREC and reviewer for JAIR, TKDE, TKDD, JMLR, TACL journals.                      |

## RESEARCH GRANTS

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Total research income so far GBP 4,102,357.

1. **Illuminating the dark metabolome: de novo identification of small molecules from their mass spectra using transformer-based deep learning**, Biotechnology and Biological Science Research Council (BBSRC), Co-I, (GBP 833,502), 2024-2027.
2. **DynAIRx – AIs for Dynamic prescribing optimisation and care integration in multimorbidity** (NIHR), Co-I, (GBP 1,204K), 2022-2024.
3. **Procedural Natural Language Inference** (Cookpad), PI, (GBP 48k), 2017-2019.
4. **Legal Advisor Dialogue Engine** (LexSnap), PI, (GBP 12k), 2017-2018.
5. **Algorithm Design for Automatic Classification of Transactions into a Taxonomy** (Rosslyn Data Technologies), PI, (GBP 5k), 2017-2018.
6. **Track Analytics For Effective Triage of Wide Area**, Defence Science Technology Laboratory (DSTL), Co-I (GBP 243k), 2017-2019.
7. **Digital Legal Assistant**, Knowledge Transfer Partnership (Innovate UK), PI (GBP 492k), 2017-2020.
8. **WEB-RADR: Recognising Adverse Drug Reactions**, (European Commission) Innovative Medicine Initiative, Co-I (GBP 471k), 2014-2017.
9. **I knew that relation from news**, Innovation Voucher Scheme, University of Liverpool, PI (GBP 5k), 2015-2016.
10. **The Revierview Law Contract Map Project**, Knowledge Transfer Partnership (Innovate UK), Co-I (GBP 269k), 2015-2018.
11. **Resolving Relational Ambiguity between Entities on the Web**, Microsoft Research (MSR) CORE-9 Research Grant, PI, (GBP 10K), 2013-2015.

12. **Domain Adaptation for Semantic Relation Extraction**, Japanese Society for the Promotion of Science (JSPS), Research Grant for Young Researchers (B). PI (GBP 20K), 2012–2015.
13. **Cross-Language Relational Search**, Microsoft Research (MSR) CORE-7 Research Grant, PI, (GBP 20K), 2011–2012.
14. **Developing a Cross-Language Web Search Engine**, Information Technology Promotion Agency of Japan (IPA) grant for explorative software development (Mito Project), PI, (GBP 26.5K), 2010–2011.
15. **A Latent Relational Search Engine**, Google Research Award, Co-I, (GBP 18.6K), 2010–2011.
16. **Developing a Relational Search Engine to Retrieve Semantic Relations between Entities**, Japanese Society for the Promotion of Science (JSPS) research grant, PI. (GBP 29.8K), 2010–2012.
17. **Research grant for overseas visiting scholars**, Global Centre of Excellence (GCOE), Japan. PI. (GBP 9.7K), April 2011–June 2011.
18. **Extracting Attributes for Entities using Web Data**, Global Centre of Excellence (GCOE), Japan. PI. (GBP 7.5K), 2010–2011.
19. **Learning to Rank Entities on the Web**, Microsoft Research (MSR) CORE-6 Research Grant, Co-I, (GBP 19.4K), 2010–2011.
20. **Using Web Mining to Provide Useful Information to Drivers**, Toyota InfoTechnology Centre, Co-I, (GBP 29.8K) 2009–2012.
21. **Disambiguating Personal Names on the Web**, Japan Society for the Promotion of Science (JSPS) Research Grant PI. (GBP 29.8K), 2007–2009.
22. **Using Network Theory and Machine Learning to Structure and Represent Information Available on the Web**, Co-I, (GBP 298K), 2009–2012.

## SUPERVISION OF PHD STUDENTS

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Graduated PhD Students under my primary supervision:

1. Guanqun Cao, March 2024 (Co-supervised with Shan Luo), now Lecturer at University of West England, Bristol, UK.
2. Samantha Durdy, March 2023 (Co-supervised with Matt Rosseinsky and Micheal Gaultois).
3. Yi Zhou (Jodie), March 2023, now Lecturer at Cardiff University.
4. Micheal Abaho, Sep 2022, now PDRA at University of Liverpool.
5. James O'Neill, May 2021, now Research Scientists at Huawei.
6. Huda Hakami, May 2020, now Assistant Professor at Taif University, Saudi Arabia.
7. Xia Cui, graduated April 2020, now Lecturer at Manchester Metropolitan University, UK.
8. Alsuhaibani Mohammed, March 2020, now Assistant Professor at Qassim University, Saudi Arabia.
9. Pavithra Rajendran, graduated March 2019, now data scientist at National Health Services (NHS), UK.

10. Asif Hussain Khan, graduated March 2014, now assistant professor, University of Dhaka.
11. Leon Palafox, graduated March 2012, now postdoc at University of Arizona.
12. Liu Shu, graduated March 2011, now engineer at Microsoft.
13. Makoto Tanji, graduated March 2011, now engineer at Wantedly.
14. Akio Watanabe, graduated March 2012, now engineer at CyberAgent.
15. Nguyen Tuan Duc, graduated March 2011, now Vice President at Alt+.
16. Hugo Hernault, graduated March 2011, now engineer at Barclays.
17. Abdullah Alsheri, graduated June 2017, now lecturer at Saudi Arabia.

## PHD EXAMINATIONS:

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1. Francesco Periti, University of Milan, Italy, October, 2024.
2. Silvia Severini, University of Munich, Germany, December, 2022.
3. Hicham El Boukkouri, LISN, CNRS, France, November, 2021.
4. Rana Alshaikh, University of Cardiff, May 2021.
5. Abiola Obamuyide, University of Sheffield, June 2020.
6. Alexander Phillips, University of Liverpool, January 2020.
7. Mohammed Al-Zeyadi, University of Liverpool, July 2018.
8. Bastian Broecker, University of Liverpool, April 2018.
9. Fatima Abdullahi, University of Liverpool, May 2016.
10. Liyung Gong, University of Liverpool, November, 2014.
11. Tacao Renevey Francisco, University of Tokyo, March, 2013.
12. Mamdouh Farouk Mohamed, University of Tokyo, March, 2012.
13. Haibo Li, University of Tokyo, March 2011.
14. Alena Neviarouskaya, University of Tokyo, March 2011.

## SELECTED KEYNOTES/INVITED TALKS

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1. Keynote at the \*SEM 2023, Toronto, Canada, 2023.
2. Invited talk at the Open Data Science Conference (ODSC), London, 2023.
3. Invited talk at the Digital Theme UK-Ukraine Twinning Initiative, Liverpool, 2023.
4. Invited talk at the World Human Resource Conference, Colombo, 2022.
5. Invited talk at the Open Data Science Conference (ODSC), London, 2022.
6. Keynote at National Human Resource Conference, Colombo, 2018.
7. Invited talk at International Society for Pharmacovigilance, 2017.
8. Invited talk at Microsoft Research Beijing Lab, 2013.
9. Keynote at Information-Based Induction Sciences (IBIS) Conference, 2011.
10. Invited talk at Google Mountain View Lab, 2011.

11. Invited talk at Microsoft Research Seattle Lab, 2010.
12. Keynote at First Japanese Web Symposium, 2009.

## PATENTS

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1. Query Anonymisation via Semantic Decomposition, Japanese patent (filed 2018.06 and patent-pending).
2. A Method for Extracting the Semantic Relations that exist between two Entities from a Text Corpus, Japanese patent no: 2010-096551, 2010.
3. A Relational Search System, Japanese patent no: 2009-275762, 2009.

## FACULTY IMPACT CASE RECORDS

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1. AI & Law Impact case covering KTP projects with Riverview Law, Fletchers solicitors, and consultancy project with LexSnap.
2. Impact case for the monitoring adverse reactions of drugs from social media for pharmacovigilance (WEB-RADR project)

## INDUSTRIAL COLLABORATIONS/CONSULTANCIES

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|                 |                              |
|-----------------|------------------------------|
| 2018.10-CURRENT | Amazon Scholar               |
| 2018.04-2019.01 | NLP consultant, LexSnap Ltd. |
| 2017.08-2022.08 | Senior Fellow, Cookpad Ltd.  |

## PUBLICATIONS

I have published over 200 papers in top international venues in Natural Language Processing, Machine Learning, Data Mining, Artificial Intelligence, and Social Media Analysis. My papers have been cited 6264 times with an  $h$ -index of 41 and  $i10$ -index of 106. For a full list of my publications and metrics see [Google Scholar Profile](#).

## REFERRED JOURNAL PAPERS

- [1] Yang Zhao, Danushka Bollegala, Shunsuke Hirose, Yingzi Jin, and Tomotake Koza. Community knowledge graph abstraction for enhanced link prediction: A study on PubMed knowledge graph. *J. Biomed. Inform.*, 158(104725):104725, October 2024.
- [2] Samantha Durdy, Michael W. Gaultois, Vladimir Gusev, Danushka Bollegala, and Matthew J. Rosseinsky. Metrics for quantifying isotropy in high dimensional unsupervised clustering tasks in a materials context, 2023.
- [3] Katie Atkinson and Danushka Bollegala. AI for patent essentiality review. *SSRN Electronic Journal*, 2022.
- [4] Samantha Durdy, Michael Gaultois, Vladimir Gusev, Danushka Bollegala, and Matthew J. Rosseinsky. Random projections and kernelised leave one cluster out cross-validation: Universal baselines and evaluation tools for supervised machine learning for materials properties. *Digital Discovery*, 2022.
- [5] Tadashi Tsubota, Danushka Bollegala, Yang Zhao, Yingzi Jin, and Tomotake Koza. Improvement of intervention information detection for automated clinical literature screening during systematic review. *Journal of Biomedical Informatics*, page 104185, 2022.
- [6] Lauren E Walker, Aseel S Abuzour, Danushka Bollegala, Andrew Clegg, Mark Gabbay, Alan Griffiths, Cecil Kullu, Gary Leeming, Frances S Mair, Simon Maskell, Samuel Relton, Roy A Ruddle, Eduard Shantsila, Matthew Sperrin, Tjeerd Van Staa, Alan Woodall, and Iain Buchan. The dynairx project protocol: Artificial intelligence for dynamic prescribing optimisation and care integration in multimorbidity. *Journal of Multimorbidity and Comorbidity*, 12:26335565221145493, 2022.
- [7] Micheal Abaho, Danushka Bollegala, Paula Williamson, and Susanna Dodd. Assessment of contextualised representations in detecting outcome phrases in clinical trials. *European Journal for Biomedical Informatics*, 17(9), September 2021.
- [8] Masaru Isonuma, Danushka Bollegala, Junichiro Mori, and Ichiro Sakata. Unsupervised abstractive opinion summarization by generating sentences with tree-structured topic guidance. *Transactions of the Association for Computational Linguistics (TACL)*, 2021.
- [9] Katie Atkinson, Trevor Bench-Capon, and Danushka Bollegala. Explanation in ai and law: Past, present and future. *Artificial Intelligence*, 289:103387, 2020.
- [10] Yash Khemchandani, Steve O’Hagan, Soumitra Samanta, Neil Swainston, Timothy J Roberts, Danushka Bollegala, and Douglas Kell. Deepgraphmolgen, a multi-objective, computational strategy for generating molecules with desirable properties: a graph convolution and reinforcement learning approach. *Journal of Cheminformatics*, 2020.
- [11] Yi Zhou and Danushka Bollegala. Predicting the quality of translations without an oracle. In *Communications in Computer and Information Science*, pages 3–23. Springer International Publishing, 2020.



- [12] Abdullah Alshehri, Frans Coenen, and Danushka Bollegala. Behavioural biometric continuous user authentication using multivariate keystroke streams in the spectral domain. In Ana Fred, David Aveiro, Jan L. G. Dietz, Kecheng Liu, Jorge Bernardino, Ana Salgado, and Joaquim Filipe, editors, *Knowledge Discovery, Knowledge Engineering and Knowledge Management*, pages 43–66, Cham, 2019. Springer International Publishing.
- [13] Abdullah Alshehri, Frans Coenen, and Danushka Bollegala. Iterative time keystroke continuous authentication: A time series based approach. *KI - Künstliche Intelligenz*, 32(1):1–13, 2018.
- [14] Mohammed Alsuhaibani, Danushka Bollegala, Takanori Maehara, and Ken-ichi Kawarabayashi. Jointly learning word embeddings using a corpus and a knowledge base. *Plos One*, 13(3):1–26, 2018.
- [15] Danushka Bollegala, Vincent Atanasov, Takanori Maehara, and Ken-ichi Kawarabayashi. Classinet – predicting missing features for short-text classification. *ACM Transactions on Knowledge Discovery from Data (TKDD)*, pages 1–29, 2018.
- [16] Danushka Bollegala, Richard Slone, Simon Maskell, Joanne Hajne, and Munir Pirmohammed. Learning causality patterns for detecting adverse drug reactions from social media. *Journal of Medical Internet Research Public Health Surveillance*, 4(2):1–20, 2018.
- [17] Xia Cui, Noor Al-Bazzaz, Danushka Bollegala, and Frans Coenen. A comparative study of pivot selection strategies for unsupervised cross-domain sentiment classification. *The Knowledge Engineering Review*, 33:1–24, 2018.
- [18] Tomoyuki Kajiwar, Danushka Bollegala, Yuichi Yoshida, and Ken-ichi Kawarabayashi. An iterative approach for the global estimation of sentence similarity. *PLoS ONE*, 12(9):1–15, July 2017.
- [19] Danushka Bollegala. Dynamic feature scaling for online learning of binary classifiers. *Knowledge-Based Systems*, 129:97–105, 2017.
- [20] Danushka Bollegala, Kohei Hayashi, and Ken-ichi Kawarabayashi. Learning linear transformations between counting-based and prediction-based word embeddings. *PLoS ONE*, 12(9):1–21, 2017.
- [21] Huda Hakami and Danushka Bollegala. Compositional approaches for representing relations between words: A comparative study. *Knowledge-Based Systems*, 136C:172–182, 2017.
- [22] Danushka Bollegala, Georgios Kontonatsios, and Sophia Ananiadou. A cross-lingual similarity measure for detecting biomedical term translations. *PLOS ONE*, 10(6):1–28, 06 2015.
- [23] Danushka Bollegala, Tingting Mu, and Yannis Goulermas. Cross-domain sentiment classification using sentiment sensitive embeddings. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 28(2):398–410, 2015.
- [24] Hakami Huda and Danushka Bollegala. A classification approach for detecting cross-lingual biomedical term translations. *Natural Language Engineering*, 1(1469-8110):1–21, 2015.
- [25] Nozomi Nori, Danushka Bollegala, and Mitsuru Ishizuka. Interest prediction via user’s actions on social media. *Transactions of the Japanese Society for Artificial Intelligence*, pages 168–176, 2015.

- [26] Nozomi Nori, Danushka Bollegala, and Hisashi Kashima. Simultaneous higher-order relation prediction via collective incidence matrix embedding. *Transactions of the Japanese Society for Artificial Intelligence*, pages 459–465, 2015.
- [27] Hiroyuki Sato, Yoshihiko Hasegawa, Danushka Bollegala, and Hitoshi Iba. Improved sampling using loopy belief propagation for probabilistic model building genetic programming. *Swarm and Evolutionary Computation*, pages 1–8, 2015.
- [28] Richard Sloane, Orod Osanlou, David Lewis, Danushka Bollegala, Simon Maskell, and Munir Pirmohamed. Social media and pharmacovigilance: A review of the opportunities and challenges. *British Journal of Clinical Pharmacology*, 80(4):910 – 920, 2015.
- [29] Nozomi Nori, Danushka Bollegala, and Hisashi Kashima. A dimension reduction approach to multinomial relation prediction. *Transactions of the Japanese Society for Artificial Intelligence*, pages 168–176, 2014.
- [30] Danushka Bollegala. Deep learning for natural language processing. *Journal of the Japanese Society for Artificial Intelligence*, pages 195–203, 2013.
- [31] Danushka Bollegala, Tomokazu Goto, Nguyen Tuan Duc, and Mitsuru Ishizuka. Improving relational similarity measurement using symmetries in proportional word analogies. *Information Processing and Management*, 49(1):355 – 369, 2013.
- [32] Danushka Bollegala, Yutaka Matsuo, and Mitsuru Ishizuka. Minimally supervised novel relation extraction using latent relational mapping. *IEEE Transactions on Knowledge and Data Engineering*, 25(2):419 – 432, 2013.
- [33] Danushka Bollegala and Ekaterina Shutova. Metaphor interpretation using paraphrases extracted from the web. *PLoS ONE*, 8(9):1–10, 2013.
- [34] Danushka Bollegala, David Weir, and John Carroll. Cross-domain sentiment classification using a sentiment sensitive thesaurus. *IEEE Transactions on Knowledge and Data Engineering*, 25(8):1719 – 1731, 2013.
- [35] Muhammad Asif Hossain Khan, Danushka Bollegala, Guangwen Li, and Kaoru Sezaki. Delineating real-time events by identifying relevant tweets with popular discussion points. *ASE Human Journal*, 2(3):136 – 150, 2013.
- [36] Ken-ichi Yokote, Danushka Bollegala, and Mitsuru Ishizuka. Jointly learning similarity transformations for textual entailment. *Transactions of of the Japanese Society for Artificial Intelligence*, pages 220–229, 2013.
- [37] Danushka Bollegala, Yutaka Matsuo, and Mitsuru Ishizuka. Automatic annotation of ambiguous personal names on the web. *Computational Intelligence*, 28(3):398 – 425, 2012.
- [38] Danushka Bollegala, Yutaka Matsuo, and Mitsuru Ishizuka. Measuring the degree of synonymy between words using relational similarity between word pairs as a proxy. *Institute of Electronics, Information and Communication Engineers (IEICE) Transactions on Information Systems*, pages 2116–2123, 2012.
- [39] Danushka Bollegala, Naoaki Okazaki, and Mitsuru Ishizuka. A preference learning approach to sentence ordering for multi-document summarization. *Information Sciences*, 217:78 – 95, 2012.
- [40] Danushka Bollegala, Naoki Tani, and Mitsuru Ishizuka. Improving the accuracy of attribute extraction using the relatedness between attribute values. *Transactions of the Japanese Society for Artificial Intelligence*, pages 245–252, 2012.

- [41] Nguyen Tuan Duc, Danushka Bollegala, and Mitsuru Ishizuka. Cross-language latent relational search between japanese and english languages using a web corpus. *ACM Transactions on Asian Language Processing (TALIP)*, 11(3):1 – 33, 2012.
- [42] Danushka Bollegala, Yutaka Matsuo, and Mitsuru Ishizuka. Automatic discovery of personal name aliases from the web. *IEEE Transactions on Knowledge and Data Engineering*, 23(6):831 – 844, July 2011.
- [43] Danushka Bollegala, Yutaka Matsuo, and Mitsuru Ishizuka. A web search engine-based approach to measure semantic similarity between words. *IEEE Transactions on Knowledge and Data Engineering (TKDE)*, 23(7):977–990, July 2011.
- [44] Danushka Bollegala, Yutaka Matsuo, and Mitsuru Ishizuka. A supervised classification approach for measuring similarity between word pairs. *Transactions of the Institute of Electronics, Information and Communication Engineers (IEICE)*, E94-D(11):2227–2233, 2011.
- [45] Nguyen Tuan Duc, Danushka Bollegala, and Mitsuru Ishizuka. Exploiting relational similarity between entity pairs for latent relational search. *Transactions of the Information Processing Society of Japan*, 52(4):1790–1802, 2011.
- [46] Nguyen Tuan Duc, Danushka Bollegala, and Mitsuru Ishizuka. Relation representation and indexing method for fast and high precision latent relational search engine. *Special issue of the Transactions of the Japanese Society for Artificial Intelligence*, 26(2):307–312, 2011.
- [47] Tomokazu Goto, Nguyen Tuan Duc, Danushka Bollegala, and Mitsuru Ishizuka. Improving relational search performance using relational symmetries and predictors. *Transactions of the Japanese Society for Artificial Intelligence*, 26(6):649–656, 2011.
- [48] Wataru Sunayama, Yasufumi Takama, Danushka Bollegala, Yoko Nishihara, Hidekazu Tokunaga, Mineo Kushima, and Mitsunori Matsushita. Total environment for text data mining. *Transactions of the Japanese Society for Artificial Intelligence*, 26(4):483–493, 2011.
- [49] Danushka Bollegala, Naoaki Okazaki, and Mitsuru Ishizuka. A bottom-up approach to sentence ordering for multi-document summarization. *Information Processing and Management*, 46(1):89 – 109, 2010.
- [50] Keigo Watanabe, Danushka Bollegala, Yutaka Matsuo, and Mitsuru Ishizuka. Automatic extraction of related terms using web search engines. *Journal of the Japan Society for Fuzzy Theory and Intelligent Informatics*, 23(4):483–493, 2010.
- [51] D. BOLLEGALA, N. OKAZAKI, and M. ISHIZUKA. Agglomerative clustering based approach to sentence ordering for multi-document summarization. *IEIC Technical Report (Institute of Electronics, Information and Communication Engineers)*, 105(594):13–18, 2006.

## REFERRED CONFERENCE PAPERS

- [52] Junjie Chen, Xiangheng He, Yusuke Miyao, and Danushka Bollegala. Improving unsupervised constituency parsing via maximizing semantic information. In *The Thirteenth International Conference on Learning Representations*, 2025.
- [53] Taichi Aida and Danushka Bollegala. A semantic distance metric learning approach for lexical semantic change detection. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics ACL 2024*, pages 7570–7584, Bangkok, Thailand and virtual meeting, August 2024. Association for Computational Linguistics.

- [54] Junjie Chen, Xiangheng He, Danushka Bollegala, and Yusuke Miyao. Unsupervised parsing by searching for frequent word sequences among sentences with equivalent predicate-argument structures. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar, editors, *Findings of the Association for Computational Linguistics ACL 2024*, pages 3760–3772, Bangkok, Thailand and virtual meeting, August 2024. Association for Computational Linguistics.
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## TEACHING EXPERIENCE

**COMP 527: Data Mining and Visualisation** I have been teaching COMP 527 continuously since 2014. It is a master-level module and a compulsory module for the MSc programme in Big Data and High Performance Computing. This module carries 15 credits and is taught in the second semester. Since I have taken over this module, I have re-written the module specifications and have added more practical elements such as the introduction of Python-based programming course work and lab assignments. Initially, video recordings of the lectures were made available on a dedicated YouTube channel, and later when the university introduced the video streaming platform, the video captures from the lectures were made available at [stream.liv.ac.uk](http://stream.liv.ac.uk). The exam structure was also changed to introduce more problem solving-type questions and this effort has been repeatedly praised by the external examiners. This module is a popular choice among the computer science PGT cohort and the number of students taking this module has increased significantly over the year from 16 in 2014 to 48 in 2018. It is the module with the largest number of student registration among all PGT modules in the department. The module is also taken by a large number of PhD students from various CDT programmes such as the Data Analytics and Society CDT, Risk CDT and Physics CDT because data science has become an integrated component in many research fields not limiting to computer science. The student feedback for COMP 527 this year was extremely positive and all questions in the student evaluation received a high average rating of 4.1 or above.

**COMP 212: Distributed Systems** This is a second year 15 credit undergraduate model that is optional for all computer science students at the Department of Computer Science, University of Liverpool. The module covers both theoretical as well as practical aspects of distributed computing. In addition to the written exam, there are two programming assignments that must be implemented using the Java programming language, testing the students' understanding of the algorithms in distributed systems. I taught this module for three consecutive years during the period from 2014 to 2016. On average, 40 students were registered for this module during that period. The student feedback for COMP 212 has been positive during that period. According to the departmental policy, undergraduate admission tutor is given a lower teaching load, and as a result I discontinued teaching COMP 212 from 2017.

**C Programming (University of Tokyo)** This is a second year compulsory module for all students in the Department of Information and Communication Engineering at the University of Tokyo, Japan. Techniques for optimising programmes written for lower-level hardware devices are covered in this module. This is an intensive programming-oriented module with weekly assignments. On average 150 students take this module and it is the responsibility of the module coordinator to assess each submitted assignment and provide detailed feedback on a weekly basis. This module enabled me to gain valuable experience related to teaching large cohorts. In particular, I developed a machine learning-based automatic programme evaluation system that can automatically grade the student assignments and highlight common mistakes. This enabled students to submit their assignments many times as they wish before the deadline and obtain real-time feedback. This also encouraged students to submit the assignments well before the deadlines. After the deadline has passed, I personally verified the mistakes detected by the tool and provided an annotated feedback to the students. It would have been extremely difficult and time consuming to conduct this module without this innovation.